INCH-POUND
MIL-DTL-32687/1
w/Amendment 3
29 May 2026
SUPERSEDING
MIL-DTL-32687/1
w/Amendment 2
18 July 2025

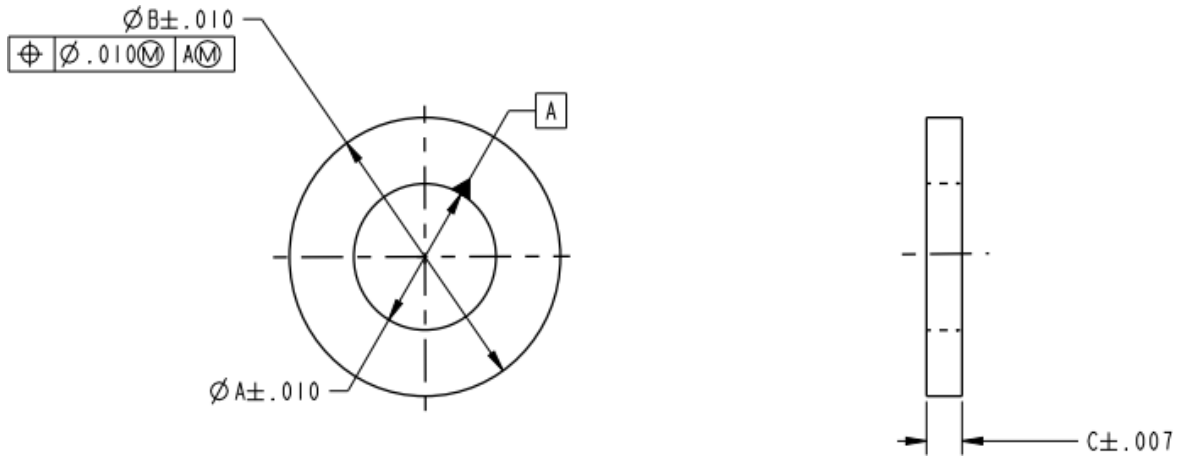
DETAIL SPECIFICATION SHEET

WASHER, FLAT, ALLOY OR CARBON STEEL, ROUND, HARDENED, ZINC-NICKEL PLATED

(Similar to 19207-10888410, 19207-10910174, 19207-12387327, 19207-12569613 and 19207-12662533)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32687.



NOTES:
1. Dimensions are in inches.
2. For M32687/1-50, limit deviation to specified diameter (A) is +0.010.
3. If stamping is used to manufacture washers, tolerance for dimensions A and B shall be + 0.040/ -0.010.
4. All edges and corners shall be free from burrs.
5. Unless otherwise specified, edges shall be broken R 0.02+ 0.02/ - 0.00.

FIGURE 1. Washer, flat, hardened.

Comments, suggestions, or questions on this document should be addressed to U.S. Army DEVCOM Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using ASSIST at https://assist.dla.mil.

AMSC N/A FSC 5310

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REQUIREMENTS:

1 Steel composition. Unless otherwise specified, washers shall be made from one of the following materials:

Steel, alloy, strip, sheet or plate, grade or designation 4130, 4140, or 8640, spec ASTM A506 or ASTM A829/A829M
Steel, alloy, bar, grade 4130, 4140, 8640, or 8740, spec ASTM A108 or ASTM A322
Steel, alloy, strip, sheet or plate, 4130, spec SAE AMS6350, SAE AMS6351 or SAE AMS6345

For only M32687/1-50 - Steel, carbon strip, cold-rolled, designation 1045 thru 1095, spec ASTM A684/A684M.

2 Hardness. Washers shall be quenched and tempered to HRC 33 through 38 (or equivalent hardness scale).

3 Finish. Zinc-nickel plating per MIL-PRF-32647, Type K20, or
Zinc-nickel plating per ASTM F1941/F1941M, Fe/Zn-Ni 8BNS.

4 Part or Identifying Number (PIN). The PIN consists of the following:

M 32687/1 - 1

- 1 — Dash number from Table I indicating size/dimensions (cross-referenced to Figure 1).
- 32687/1 — Specification Sheet number
- M — M prefix

Example: M32687/1-1 is a Flat Washer - Nominal 0.250 0.281 ID, 0.500 OD, and 0.089 inch thickness with Zinc-Nickel, Type K20 Plating per MIL-PRF-32647, or Zinc-Nickel plating per ASTM F1941/F1941M, Fe/Zn-Ni 8BNS. (See Table II for cross-reference from part numbers 19207-10888410, 19207-10910174, 19207-12387327, 19207-12569613 and 19207-12662533 to PIN.)

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TABLE I. Dimensions.

PIN	Nom Size	A	B	C
M32687/1-1	0.250	0.281	0.500	0.089
M32687/1-2	0.312	0.343	0.625	0.104
M32687/1-3	0.375	0.406	0.750	0.119
M32687/1-4	0.437	0.458	0.875	0.119
M32687/1-5	0.500	0.531	1.000	0.134
M32687/1-6	0.562	0.594	1.125	0.134
M32687/1-7	0.625	0.656	1.250	0.164
M32687/1-8	0.750	0.812	1.500	0.179
M32687/1-9	0.875	0.938	1.781	0.194
M32687/1-10	1.000	1.062	2.031	0.209
M32687/1-11	1.125	1.250	2.312	0.218
M32687/1-12	1.250	1.375	2.500	0.218
M32687/1-13	1.375	1.500	2.781	0.224
M32687/1-14	1.500	1.625	3.093	0.224
M32687/1-15	0.750	0.812	2.000	0.179
M32687/1-16	0.375	0.406	1.125	0.199
M32687/1-17	0.312	0.343	0.625	0.062
M32687/1-18	0.625	0.656	1.250	0.250
M32687/1-19	0.312	0.343	0.875	0.119
M32687/1-20	0.250	0.281	1.000	0.089
M32687/1-21	0.625	0.656	1.000	0.250
M32687/1-22	0.375	0.406	0.875	0.134
M32687/1-23	0.750	0.781	1.687	0.250
M32687/1-24	0.750	0.781	1.500	0.093

PIN	Nom Size	A	B	C
M32687/1-25	0.656	0.656	2.500	0.125
M32687/1-26	0.437	0.468	0.750	0.119
M32687/1-27	0.312	0.343	0.531	0.104
M32687/1-28	0.500	0.531	2.000	0.209
M32687/1-29	1.000	1.020	1.720	0.060
M32687/1-30	0.500	0.531	0.875	0.089
M32687/1-31	0.625	0.656	1.000	0.119
M32687/1-32	1.625	1.687	3.000	0.224
M32687/1-33	0.375	0.406	0.700	0.158
M32687/1-34	0.250	0.281	0.625	0.089
M32687/1-35	0.437	0.468	0.875	0.250
M32687/1-36	0.750	0.760	1.500	0.179
M32687/1-37	0.500	0.531	0.875	0.060
M32687/1-38	0.625	0.656	1.750	0.250
M32687/1-39	0.750	0.781	4.188	0.188
M32687/1-40	0.500	0.531	1.500	0.060
M32687/1-41	0.750	0.781	1.500	0.060
M32687/1-42	0.750	0.781	1.500	0.125
M32687/1-43	0.500	0.531	0.875	0.127
M32687/1-44	0.625	0.656	1.500	0.089
M32687/1-45	1.875	1.875	3.000	0.170
M32687/1-46	0.406	0.406	1.750	0.125
M32687/1-47	0.375	0.406	0.875	0.250
M32687/1-48	0.250	0.281	2.000	0.089
M32687/1-50	0.750	0.812	1.500	0.125

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TABLE II. Reference and Zn-Ni new part numbers. /1, /3

Reference Part Number (Zn-HC) /2, /4	Reference Part Number (Zn-HC) /2, /4	Reference Part Number (Zn-TCP) /4	New Part Number (Zn-Ni) /4
10910174-1	12387327-1	12569613-1	M32687/1-1
10910174-2	12387327-2	12569613-2	M32687/1-2
10910174-3	12387327-3	12569613-3	M32687/1-3
10910174-4	12387327-4	12569613-4	M32687/1-4
10910174-5	12387327-5	12569613-5	M32687/1-5
10910174-6	12387327-6	12569613-6	M32687/1-6
10910174-7	12387327-7	12569613-7	M32687/1-7
10910174-8	12387327-8	12569613-8	M32687/1-8
10910174-9	12387327-9	12569613-9	M32687/1-9
10910174-10	12387327-10	12569613-10	M32687/1-10
10910174-11	12387327-11	12569613-11	M32687/1-11
10910174-12	12387327-12	12569613-12	M32687/1-12
10910174-13	12387327-13	12569613-13	M32687/1-13
10910174-14	12387327-14	12569613-14	M32687/1-14
10910174-15	12387327-15	12569613-15	M32687/1-15
10910174-16	12387327-16	12569613-16	M32687/1-16
10910174-17	12387327-17	12569613-17	M32687/1-17
10910174-18	12387327-18	12569613-18	M32687/1-18
10910174-19	12387327-19	12569613-19	M32687/1-19
10910174-20	12387327-20	12569613-20	M32687/1-20
10910174-21	12387327-21	12569613-21	M32687/1-21
10910174-22	12387327-22	12569613-22	M32687/1-22
10910174-23	12387327-23	12569613-23	M32687/1-23
10910174-24	12387327-24	12569613-24	M32687/1-24

Reference Part Number (Zn-HC) /2, /4	Reference Part Number (Zn-HC) /2, /4	Reference Part Number (Zn-TCP) /4	New Part Number (Zn-Ni) /4
10910174-25	12387327-25	12569613-25	M32687/1-25
10910174-26	12387327-26	12569613-26	M32687/1-26
10910174-27	12387327-27	12569613-27	M32687/1-27
10910174-28	12387327-28	12569613-28	M32687/1-28
10910174-29	12387327-29	12569613-29	M32687/1-29
10910174-30	12387327-30	12569613-30	M32687/1-30
10910174-31	12387327-31	12569613-31	M32687/1-31
10910174-32	12387327-32	12569613-32	M32687/1-32
10910174-33	12387327-33	12569613-33	M32687/1-33
10910174-34	12387327-34	12569613-34	M32687/1-34
10910174-35	12387327-35	12569613-35	M32687/1-35
10910174-36	12387327-36	12569613-36	M32687/1-36
10910174-37	12387327-37	12569613-37	M32687/1-37
10910174-38	12387327-38	12569613-38	M32687/1-38
10910174-39	12387327-39	12569613-39	M32687/1-39
10910174-40	12387327-40	12569613-40	M32687/1-40
10910174-41	12387327-41	12569613-41	M32687/1-41
10910174-42	12387327-42	12569613-42	M32687/1-42
10910174-43	12387327-43	12569613-43	M32687/1-43
10910174-44	12387327-44	12569613-44	M32687/1-44
10910174-45	12387327-45	12569613-45	M32687/1-45
10910174-46	12387327-46	12569613-46	M32687/1-46
	12387327-47		M32687/1-47

/1 The difference between the parts from the reference and new parts is the finish.
/2 Reference part numbers that are zinc plated with hexavalent chromium are not recommended for new design on Army ground systems.
/3 Use new part numbers in place of reference part numbers with design authority approval.
/4 Zn-Ni: Zinc-Nickel plated
Zn-HC: Zinc plated with hexavalent chromium
Zn-TCP: Zinc plated with trivalent chromium process (passivation)

Table III. Additional reference and Zn-Ni new part numbers. /1, /3

Reference Part Number No finish	Reference Part Number (Cd-HC) /2, /4	New Part Number (Zn-Ni) /4
12662533-47		M32687/1-47
12662533-48		M32687/1-48
	10888410	M32687/1-50

/1 The difference between the parts from the reference and new parts is the finish.
/2 Reference part numbers that are cadmium plated are not recommended for new design on Army ground systems.
/3 Use new part numbers in place of reference part numbers with design authority approval.
/4 Zn-Ni: Zinc-Nickel plated
Cd-HC: Cadmium plated with hexavalent chromium

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CONCLUDING MATERIAL

Custodians:
Army – AT
Navy – AS
Air Force – 70

Preparing Activity:
Army – AT

(Project 5310-2026-003)

Review Activities:
Army – AR, AV, CR4, MI
Navy – OS, SH
Air Force – 11, 20

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